Nicolaus Copernicus in the sixteenth century that we do not occupy a "special place" in the cosmos, but are just at some generic spot that may as well have been chosen at random. So when we look at a galaxy a billion light-years away, and see it as it was a billion years ago, in a universe that was a billion years younger than our universe is here and now, we can be pretty confident that the conditions *here* a billion years ago would have been fairly similar. This can actually be observationally tested, to a degree. Studies of the distribution of galaxies throughout the cosmos find that the uniformity implied by the cosmological principle holds up everywhere we've looked.

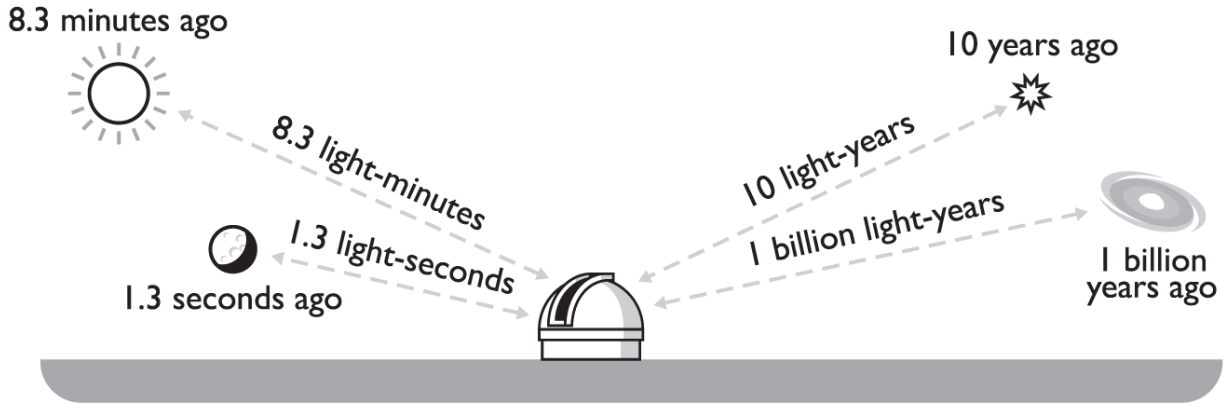


Figure 1: Light travel times. We sometimes express distances in light-seconds, light-minutes, and light-years because it makes it clear how long the light has been traveling to us, and thus how far into the past we're looking. (None of the illustrations here are to scale!)

The upshot of all this is that if we want to learn about the evolution of the universe itself, and the conditions our own Milky Way galaxy grew up in, all we have to do is *look at something far away*.

It also means that cosmology doesn't really have a well-defined concept of "now." Or rather, the "now" you experience is highly specific to you, to where you are and to what you are doing.[fn2] What does it mean to say "that supernova is going off now" if we see the light of it now, and we can watch the star explode now, but that light has been traveling for millions of years? The thing we're watching is essentially fully in the past, but the "now" for that exploded star is unobservable to us, and we won't receive any knowledge of it for millions of years, which makes it, to us, not "now," but the future.

When we think of the universe as existing in *spacetime*—a kind of all-encompassing universal grid in which space is three axes and time is a